

Surname / Van:

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FSK 176

Semester Test 1 - Semestertoets 1

Department of Physics, University of Pretoria
Departement Fisika, Universiteit van Pretoria

Date/Datum: 26/08/2017

Time/Tyd: 2.5 hours/ure

Total / Totaal : 80 max/maks

Examiner/Eksaminator: Dr JM Nel

Moderator/Moderator: RQ Odendaal

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Test register: Tear carefully on the line above and hand to the invigilator.
Toets register: Skeur versigtig op die lyn hierbo en handig dit in by die toesighouer.

FSK 176: Semester Test 1 / Semestertoets 1

26/08/2017

Surname Van					Initials Voorletters		
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Please read the following carefully. Lees die volgende noukeurig

1. Answer all the questions in the space provided on the question paper. Antwoord alle vrae in die spasie gelaat op die vraestel.
2. No pencil work will be marked. Do not use "tippex". Geen potlood-werk sal gemerk word nie. Moet nie uitwis-vloeistof ("tippex") gebruik nie.
3. If you need more space, please use the back of the pages, and clearly indicate where the question continues. As u meer spasie benodig gebruik die agterkant van die bladsye. Dui duidelik aan waar die vraag aangaan.
4. Where applicable you need to draw the necessary diagrams (e.g. free-body diagrams). Teken diagramme waar nodig (bv. vryliggaams-diagramme).
5. Your answer needs to be written in a systematic and logical fashion. You must show that you understand and can use the relevant physics. **Only substitute numerical values in the final step of calculations! A correct answer does not imply that you will receive full marks.** U antwoorde moet sistematies en logies neergeskryf word. U moet bewys dat u die nodige fisika verstaan en kan toepas. **Numeriese waardes mag slegs in die laaste stap van 'n berekening vervang word!** 'n Korrekte antwoord beteken nie dat u noodwendig vol punte sal kry nie.
6. Units must be used when numbers are substituted else marks will be subtracted. Eenhede moet gebruik word wanneer numeriese waardes vervang word, andersins sal punte afgetrek word.
7. The use of the cheat sheet is only allowed subject to the rules stated on it. Die formuleblad mag slegs gebruik word onderhewig aan die reëls daarop gedruk.
8. All exam regulations of the University of Pretoria apply, and any contravention of it will be treated in a serious manner. Alle eksamen regulasies van die Universiteit van Pretoria geld en enige oortreding daarvan sal in 'n ernstige lig beskou word.

Constants and formulas / Konstantes en formules

$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/(\text{N}\cdot\text{m}^2)$; $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$; $e = 1.602 \times 10^{-19} \text{ C}$; $k = 1.381 \times 10^{-23} \text{ J/K}$;
 $m_e = 9.109 \times 10^{-31} \text{ kg}$; $m_p = 1.673 \times 10^{-27} \text{ kg}$; $R = 8.314 \text{ J}/(\text{mol}\cdot\text{K})$; $N_A = 6.02 \times 10^{23} / \text{mol}$; $1 \text{ atm} = 101.3 \text{ kPa}$;
 $1 \text{ u} = 1.6605677 \times 10^{-27} \text{ kg}$; $c = 3 \times 10^8 \text{ m/s}$; $V_{\text{sphere/sfeer}} = \frac{4}{3}\pi r^3$; $F_{\text{gravity}} = \frac{Gm_1m_2}{r^2}$; $1 \text{ litre} = 1000 \text{ cm}^3$;
 $M_{\text{Earth}} = M_{\text{Aarde}} = 5.97 \times 10^{24} \text{ kg}$; $\rho(\text{water}) = 1 \text{ g/cm}^3$; $r_{\text{Earth}} = r_{\text{Aarde}} = 6.37 \times 10^6 \text{ m}$

1 Question 1 / Vraag 1

The Furlong, Firkin, and Fortnight System (FFF) is based around three basic measurements: the furlong for length (200 m), firkin for mass (40 kg) and fortnight for time (two weeks or 14 days). Akani Simbine has run the 100 m sprint in 9.89 s. Express his average speed during the sprint in terms of furlongs/fortnight. Show all conversion factors in your calculation. Die Furlong, Firkin en Fortnight sisteem (FFF) is gebaseer op drie metings: die furlong vir lengte (200 m), firkin vir massa (40 kg) en die fortnight vir tyd (2 weke of 14 dae). Akani Simbine het al die 100 m wedloop in 9.89 s gehardloop. Druk sy gemiddelde spoed tydens die wedloop uit in terme van "furlongs/fortnight". Wys alle omskakelingsfaktore in jou berekening.

$$v_{\text{avg}} = \frac{\text{distance}}{\text{time}} = \frac{100 \text{ m}}{9.89 \text{ s}}$$

$$= 10.11 \frac{\text{m}}{\text{s}} \times \frac{1 \text{ furlong}}{200 \text{ m}} \times \frac{3600 \text{ s}}{1 \text{ hr}} \times \frac{24 \text{ hr}}{1 \text{ day}} \times \frac{14 \text{ days}}{1 \text{ fortnight}}$$

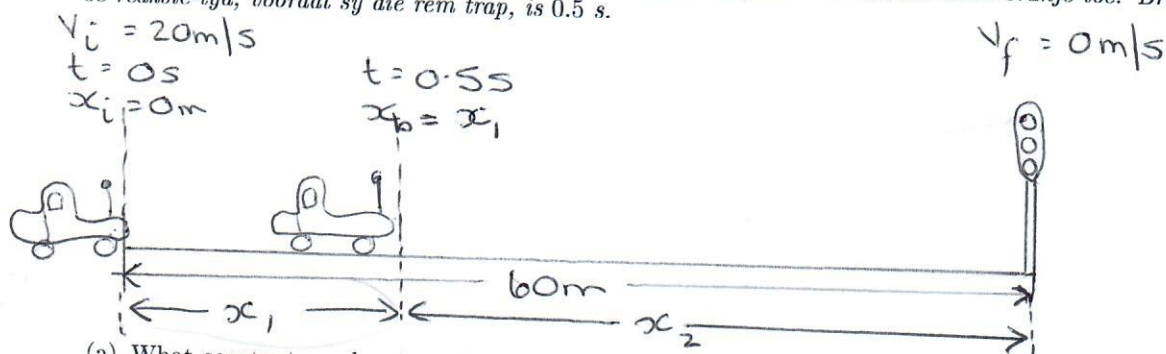
$$= 61.15 \times 10^3 \text{ furlong / fortnight}$$

2 Question 2 / Vraag 2

[8]

Brenda is travelling at 20 m/s. She is 60 m from the traffic light when she sees it turn amber (orange). Brenda's reaction time, before stepping on the brake, is 0.5 s.

Brenda beweeg teen 20 m/s. Sy is 60 m van die verkeerslig wanneer dit oorslaan oranje toe. Brenda se reaksie-tyd, voordat sy die rem trap, is 0.5 s.



- (a) What constant acceleration while braking will bring her to a stop at the traffic light?
Watter konstante versnelling sal haar tot stilstand bring by die verkeerslig?

(5)

Distance travelled before braking:

$$x_1 = v_i t = (20 \text{ m/s})(0.5 \text{ s}) = 10 \text{ m} \quad \checkmark$$

Braking distance: $x_2 = 60 \text{ m} - 10 \text{ m} = 50 \text{ m} \quad \checkmark$

Acceleration during braking:

$$v_f^2 = v_i^2 + 2ax_2 \quad \checkmark$$

$$a = \frac{v_f^2 - v_i^2}{2x_2} = \frac{0^2 - (20 \text{ m/s})^2}{2(50 \text{ m})}$$

$$= -4.0 \text{ m/s}^2 \quad \checkmark$$

- (b) From the time that the traffic light turned, how long will it take to come to a stop?
 Hoe lank sal dit neem om tot stilstand te kom vandat die verkeerslig van kleur verander het? (3)

for distance x_1 : $t_1 = 0.55$

for distance x_2 :

$$v_f = v_i + at_2$$

$$t_2 = \frac{v_f - v_i}{a} = \frac{0 - 20 \text{ m/s}}{-4.0 \text{ m/s}^2}$$

must be -4.0 m/s^2

$$t_2 = 5.0 \text{ s} \checkmark$$

$$\text{Total time: } \Delta t = t_1 + t_2 = 0.55 + 5.0 \text{ s}$$

$$\Delta t = 5.55 \checkmark$$

3 Question 3 / Vraag 3

[7]

An object starts from rest and moves along the horizontal x -direction. The acceleration of the object is defined by $a_x = \frac{dv_x}{dt}$, while the velocity of the object is defined by $v_x = \frac{dx}{dt}$. Using these definitions, clearly show, using the **calculus methods**, that the displacement of the object can be given by $x_f - x_i = v_{xi}t + \frac{1}{2}a_x t^2$. Do not leave out any steps, and clearly explain each step.

'n Voorwerp begin uit rus en beweeg horisontaal in die x -rigting. Die versnelling van die voorwerp word gedefinieer as $a_x = \frac{dv_x}{dt}$, en die snelheid van die voorwerp word gedefinieer as $v_x = \frac{dx}{dt}$. Deur gebruik te maak van hierdie vergelykings en die **differensiaalrekenene-metode**, wys duidelik dat die verplasing van die voorwerp gegee kan word deur $x_f - x_i = v_{xi}t + \frac{1}{2}a_x t^2$. Moet nie stappe uit los nie, en verduidelik elke stap.

$$a_x = dv_x/dt \quad \therefore a_x dt = dv_x$$

$$\text{Integrating both sides } \int a_x dt = \int dv_x \checkmark$$

$$c + at = v_x \checkmark$$

$$\text{But at } t=0, v_x = v_{xi} \quad \therefore c = v_{xi} \checkmark$$

$$\text{So } v_x = v_{xi} + at \checkmark \quad \text{--- (1)}$$

$$\text{Also } v_x = dx/dt \quad \therefore v_x dt = dx$$

$$\text{Integrating } \int dx = \int v_x dt \checkmark$$

Substituting v_x from (1)

$$\int dx = \int (v_{xi} + at) dt \checkmark$$

$$x = v_{xi}t + \frac{1}{2}at^2 + c' \checkmark$$

$$\text{But at } t=0, x = x_i \text{ then } c' = x_i \checkmark$$

$$\text{Then } x - x_i = v_{xi}t + \frac{1}{2}a_x t^2 \checkmark$$

4 Question 4 / Vraag 4

[11]

The position of a particle moving along the x -axis is given in centimeters by $x = 9.75 + 1.50t^3$, where t is in seconds. Calculate:

Die posisie van 'n deeltjie wat langs die x -as beweeg word in sentimeter gegee deur $x = 9.75 + 1.50t^3$, waar t in sekondes is. Bereken:

Alternative:

$$v_{ave} = \frac{x_f - x_i}{t_f - t_i}$$

- (a) the average velocity during the time interval $t = 2.00$ s to $t = 3.00$ s.
 die gemiddelde snelheid tydens die tydsinterval $t = 2.00$ s to $t = 3.00$ s.

(5)

$$v = \frac{dx}{dt} = \frac{d}{dt} (9.75 + 1.50t^3)$$

$$= 4.50t^2 \quad \checkmark$$

$$v(2s) = 18.00 \text{ cm/s} \quad \checkmark$$

$$v(3s) = 40.50 \text{ cm/s} \quad \checkmark$$

$$v_{ave} = \frac{1}{2}(v_2 + v_3) = \frac{1}{2}(18.00 \text{ cm/s} + 40.50 \text{ cm/s}) = 29.25 \text{ cm/s} \quad \checkmark$$

- (b) the acceleration at $t = 3.00$ s.
 die versnelling by $t = 3.00$ s.

(3)

$$a = \frac{dv}{dt} = \frac{d}{dt} (4.50t^2) = 9.00t \quad \checkmark$$

$$\text{At } t = 3.00 \text{ s}$$

$$a = 9.00t = 9.00(3.00 \text{ s}) \quad \checkmark$$

$$= 27.00 \text{ cm/s}^2 \quad \checkmark$$

- (c) the displacement during the time interval $t = 2.00$ s to $t = 3.00$ s.
 die verplasing tydens die tydsinterval $t = 2.00$ s to $t = 3.00$ s.

(3)

$$x(2s) = 9.75 + 1.50t^3$$

$$= 9.75 + 1.50(2.00 \text{ s})^3 = 21.75 \text{ cm} \quad \checkmark$$

$$x(3s) = 50.25 \text{ cm} \quad \checkmark$$

$$\text{Displacement: } \Delta x = x(3s) - x(2s)$$

$$= 50.25 \text{ cm} - 21.75 \text{ cm}$$

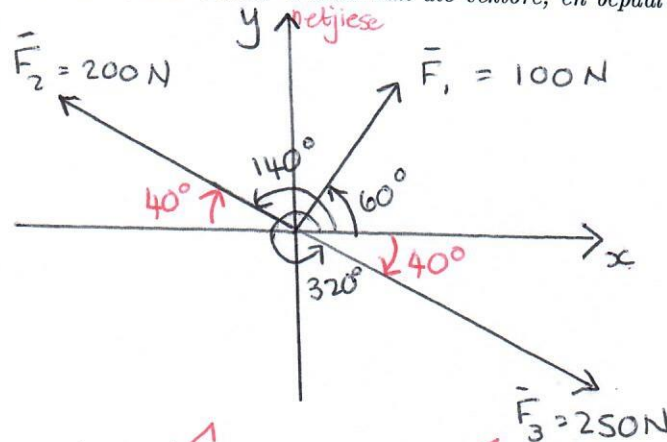
$$= 28.50 \text{ cm} \quad \checkmark$$

5 Question 5 / Vraag 5

[10]

The resultant force due to the action of four forces is $\vec{R}$, which is 100 N along the negative y -axis. Three of the forces are: 100 N directed 60° above the positive x -axis, 200 N directed 140° from the $+x$ -axis, and 250 N directed 320° from the $+x$ -axis. Draw a sketch of the vectors, and find the magnitude and direction of the fourth force.

Die resultante krag as gevolg van die aksie van vier kragte is $\vec{R}$, wat 100 N langs die negatiewe y -as is. Drie van die kragte is: 100 N gerig 60° bo die positiewe x -as, 200 N gerig 140° van die $+x$ -as, en 250 N gerig 320° van die $+x$ -as. Teken 'n skets van die vektore, en bepaal die grootte en rigting van die vierde krag.



✓ $-\frac{1}{2}$ if axes are not labelled

$$\vec{F}_1 = (100\text{ N})\cos 60^\circ \hat{i} + (100\text{ N})\sin 60^\circ \hat{j} = 50\text{ N}\hat{i} + 86.7\text{ N}\hat{j}$$

$$\vec{F}_2 = -(200\text{ N})\cos 40^\circ \hat{i} + (200\text{ N})\sin 40^\circ \hat{j} = -153\text{ N}\hat{i} + 128.6\text{ N}\hat{j}$$

$$\vec{F}_3 = (250\text{ N})\cos 40^\circ \hat{i} - (250\text{ N})\sin 40^\circ \hat{j} = 192.5\text{ N}\hat{i} - 160.7\text{ N}\hat{j}$$

$$\vec{R} = 0\hat{i} - 100\hat{j}$$

$$R_x = 0 = F_{1x} + F_{2x} + F_{3x} + F_{4x}$$

$$F_{4x} = -(F_{1x} + F_{2x} + F_{3x})$$

$$= -(50\text{ N} + (-153\text{ N}) + 192.5\text{ N})$$

$$= -90\text{ N}$$

$$R_y = -100\text{ N} = F_{1y} + F_{2y} + F_{3y} + F_{4y}$$

$$F_{4y} = -(86.7\text{ N} + 128.6\text{ N} - 160.7\text{ N}) - 100\text{ N}$$

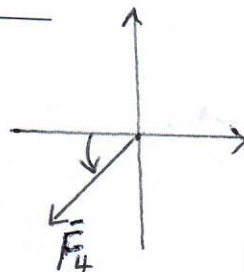
$$= -155\text{ N}$$

$$|F_4| = \sqrt{F_{4x}^2 + F_{4y}^2} = \sqrt{(-90\text{ N})^2 + (-155\text{ N})^2}$$

$$= 179\text{ N}$$

$$\theta = \tan^{-1}\left(\frac{F_{4y}}{F_{4x}}\right) = 60^\circ \text{ below } -x\text{-axis.}$$

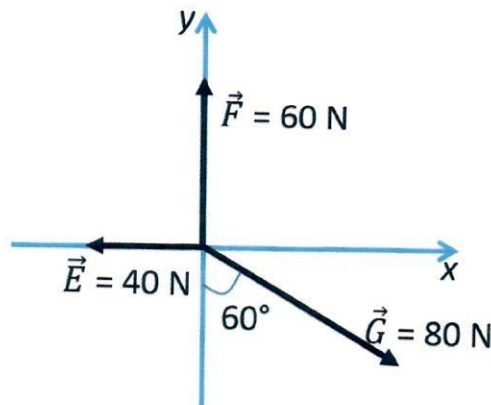
$$\text{OR } 240^\circ \text{ from } +x\text{-axis.}$$



6 Question 6 / Vraag 6

[5]

The figure shows three vectors in a plane. Find the resultant vector $\vec{E} + \vec{F} + \vec{G}$ in unit vector notation.
Die figuur dui drie vektore aan wat in 'n vlak lê. Bereken die som $\vec{E} + \vec{F} + \vec{G}$ van die vektore in eenheids-vektor notasie.



$$\begin{aligned}\vec{F} &= 0\hat{i} + 60\hat{j} \\ \vec{E} &= -40\hat{i} + 0\hat{j} \\ \vec{G} &= -80\sin 60^\circ\hat{i} - 80\cos 60^\circ\hat{j} \\ &= 69.3\hat{i} - 40\hat{j}\end{aligned}$$

$$\begin{aligned}\vec{E} + \vec{F} + \vec{G} &= (E_x + F_x + G_x)\hat{i} + (E_y + F_y + G_y)\hat{j} \\ &= (-40\text{ N} + 0 + 69.3\text{ N})\hat{i} + (0 + 60\text{ N} + (-40\text{ N}))\hat{j} \\ &= 29.3\hat{i} + 20\hat{j} \\ \text{OR } &= (29.3\hat{i} + 20\hat{j})\text{ N}\end{aligned}$$

No marks if the units are not written down.

7 Question 7 / Vraag 7

[7]

An air-traffic controller observes two aircraft on his radar screen. The first is at an altitude of 800 m, horizontal distance of 19.2 km from the airport and 25° south of west. The second aircraft is at an altitude of 1100 m, horizontal distance of 17.6 km from the airport and 20.0° south of west. Use the space below to sketch the situation and then determine the distance between the two aircraft.

'n Lugverkeerkontroleerder sien twee vliegtuie op sy radarskerm. Die eerste vliegtuig is op 'n hoogte van 800 m, op 'n horisontale afstand van 19.2 km van die lughawe en 25° suid van wes. Die tweede vliegtuig is op 'n hoogte van 1100 m, en 'n horisontale afstand van 17.6 km van die lughawe en 20.0° suid van wes. Gebruik die spasie onderaan om die situasie te skets en bereken dan die afstand tussen die twee vliegtuie.

Side View

$$P_{2H} = 1100\text{m}$$

$$P_{1H} = 800\text{m}$$

ground

With height on z-axis

N-S on the y-axis and W-E on x-axis

Positions of the two planes are:

$$\vec{P}_1 = -19.2\text{ km} \cos 25^\circ \hat{i} - 19.2\text{ km} \sin 25^\circ \hat{j} + 800\text{ m} \hat{k}$$

$$\vec{P}_2 = -17.6\text{ km} \cos 20^\circ \hat{i} - 17.6\text{ km} \sin 20^\circ \hat{j} + 1100\text{ m} \hat{k}$$

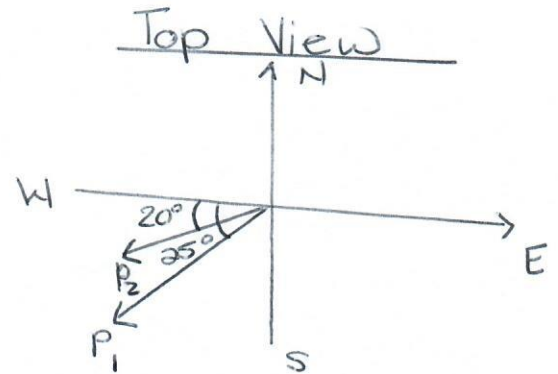
Distance between P_1 and P_2

$$\begin{aligned} \vec{P}_1 - \vec{P}_2 &= (-17.4\text{ km} - (-16.5\text{ km}))\hat{i} + (-8.11\text{ km} - (-6.02\text{ km}))\hat{j} \\ &\quad + (800\text{ m} - 1100\text{ m})\hat{k} \\ &= 900\text{ m} \hat{i} - 2090\text{ m} \hat{j} - 300\text{ m} \hat{k} \end{aligned}$$

Distance:

$$\begin{aligned} \Delta r &= \sqrt{x^2 + y^2 + z^2} \\ &= \sqrt{(900\text{ m})^2 + (-2090\text{ m})^2 + (-300\text{ m})^2} \\ &= 2295\text{ m} = 2.30\text{ km} \end{aligned}$$

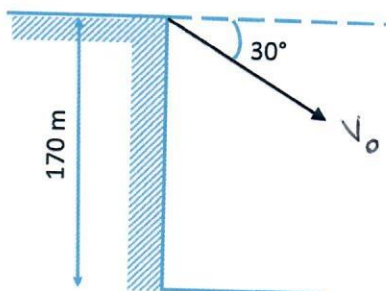
Note: signs will be different if they used $P_2 - P_1$



8 Question 8 / Vraag 8

[10]

An object is projected downward at an angle of 30° with the horizontal from the top of a building 170 m high. Its initial speed is 40 m/s.
 'n Voorwerp word afwaarts geprojekteer teen 'n hoek van 30° met die horisontaal vanaf 'n gebou wat 170 m hoog is. Die aanvanklike spoed is 40 m/s.



$$v_{0x} = v_0 \cos 30^\circ$$

$$v_{0y} = v_0 \sin 30^\circ$$

- (a) How long will it take before the object strikes the ground?
 Hoe lank neem dit voordat die voorwerp die grond tref?

$$\Delta y = v_{0y}t + \frac{1}{2}a_yt^2 = -v_0 \sin 30^\circ t + \frac{1}{2}a_yt^2 \quad (4)$$

$$-170 \text{ m} = -40 \text{ m/s} \sin 30^\circ t + \frac{1}{2}(-9.8 \text{ m/s}^2)t^2$$

$$0 = 170 \text{ m} - (20 \text{ m/s})t - (4.9 \text{ m/s}^2)t^2$$

$$\text{Using: } t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$t = 4.19 \text{ s}$$

- (b) How far from the foot of the building does the object land, **AND** at what angle does it strike the ground?
 Hoe ver van die onderkant van die gebou tref die voorwerp die grond, **EN** teen watter hoek? (6)

$$\Delta x = v_{0x}t + \frac{1}{2}a_xt^2 = v_{0x}t$$

$$= 40 \text{ m/s} \cos 30^\circ (4.19 \text{ s})$$

$$= 145 \text{ m}$$

$$v_{fx} = v_{ix} = 40 \text{ m/s} \cos 30^\circ = 34.6 \text{ m/s}$$

$$v_{fy} = v_{iy} + at = (-20 \text{ m/s}) - 9.8 \text{ m/s}^2 (4.19 \text{ s})$$

$$= -61.1 \text{ m/s}$$

$$\theta = \tan^{-1} \left(\frac{v_{fy}}{v_{fx}} \right) = \tan^{-1} \left(\frac{-61.1 \text{ m/s}}{34.6 \text{ m/s}} \right) = 60.5^\circ \text{ below horizontal}$$

9 Question 9 / Vraag 9

[9]

A ball on the end of a string is whirled around in a horizontal circle of radius 0.300 m. The plane of the circle is 1.20 m above the ground. The string breaks and the ball lands 2.00 m (horizontally) away from the point on the ground directly below the ball's location when the string breaks. Find the radial acceleration of the ball during the circular motion.

'n Bal is vas aan die punt van 'n tou en word in 'n horisontale sirkel geswaai met 'n radius van 0.300 m. Die vlak van die sirkel is 1.20 m bo die grond. Die tou breek en die bal land 2.00 m (horisontaal) weg van die punt op die grond regonder die bal se posisie toe die tou gebreek het. Bepaal die radiale versnelling van die bal tydens die sirkelbeweging.

See sketch on next page

For circular motion: $a_c = \frac{v^2}{r}$
 $v = \sqrt{a_c r}$ — (1)

In this problem $v = \sqrt{a_c r} = v_{ox}$ initial velocity of the projectile

Projectile travels 2.00m horizontally before landing and falls 1.20m

$\therefore \Delta x = v_{ox} t = 2.00 \text{ m}$ — (2)

$\Delta y = v_{oy} t + \frac{1}{2} a_y t^2 = 1.20 \text{ m}$ — (3)

$v_{oy} = 0 \text{ m/s}$ and the time taken in (2) & (3) are equal.

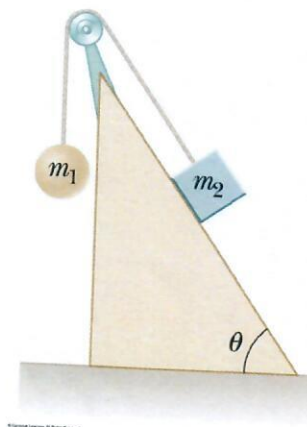
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10 Question 10 / Vraag 10

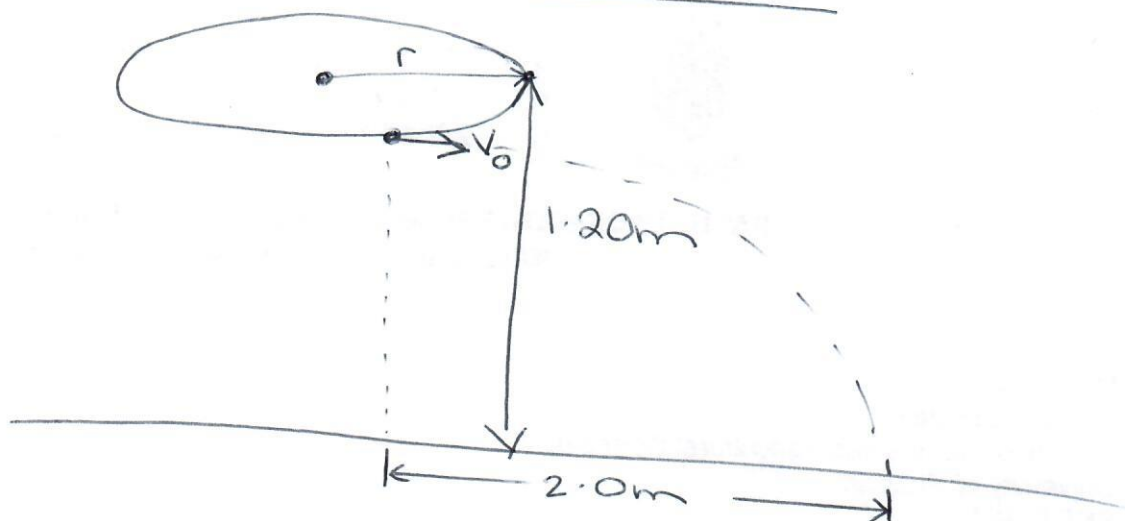
[9]

Two objects are connected by a light string that passes over a frictionless pulley with negligible mass as shown in the figure. Assume the incline is frictionless and take $m_1 = 2.00 \text{ kg}$, $m_2 = 6.00 \text{ kg}$, and $\theta = 55^\circ$.

Twee voorwerpe word aanmekaar verbind met 'n ligte tou wat oor 'n wrywinglose katrol met weglaatbare massa loop soos aangedui in die figuur. Veronderstel dat die skuinsvlak wrywingloos is en dat $m_1 = 2.00 \text{ kg}$, $m_2 = 6.00 \text{ kg}$, en $\theta = 55^\circ$.



Question 9 continued



$$v = \sqrt{a_c r} = v_{0x} \quad (1)$$

$$\Delta x = v_{0x} t = 2.00\text{m} \quad (2)$$

$$\Delta y = \underbrace{v_{0y}}_0 t + \frac{1}{2} a_y t^2 = -1.20\text{m} \quad (3)$$

$$\text{From } (2) : t = \frac{\Delta x}{v_{0x}} \quad \checkmark \quad (4)$$

④ and ① in ③ :

$$\Delta y = \underbrace{v_{0y}}_0 t + \frac{1}{2} a_y \left(\frac{\Delta x}{\sqrt{a_c r}} \right)^2 = \frac{1}{2} a_y \frac{\Delta x^2}{a_c r} \quad \checkmark$$

$$a_c = \frac{1}{2} a_y \frac{\Delta x^2}{\Delta y r} = \frac{1}{2} (-9.8\text{m/s}^2) \frac{(2.00\text{m})^2}{(-1.20\text{m})(0.300\text{m})} \quad \checkmark$$

$$a_c = 54.4\text{m/s}^2 \quad \checkmark$$

Alternative to Q9.

When string breaks:

$$\Delta y = v_{yi}t + \frac{1}{2}at^2$$

$$= 0 + \frac{1}{2}at^2$$

$$t = \sqrt{\frac{2\Delta y}{a}} = \sqrt{\frac{2(-1.20\text{m})}{-9.8\text{m/s}^2}}$$

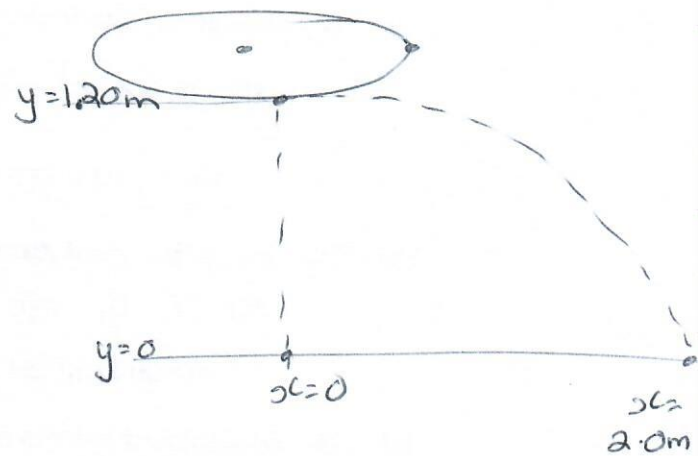
$$= 0.49\text{s}$$

$$\Delta x = v_{xi}t + \frac{1}{2}a_xt^2$$

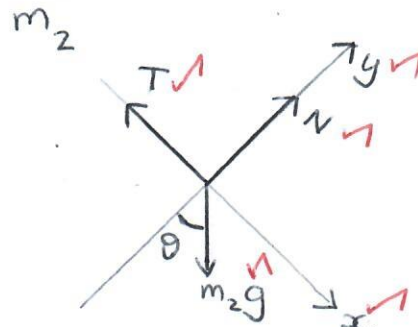
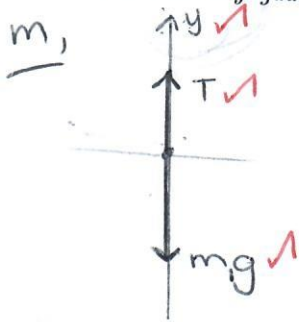
$$v_{xi} = \frac{\Delta x}{t} = \frac{2.0\text{m}}{0.49\text{s}}$$

$$v_{xi} = 4.08\text{ m/s}$$

$$a_c = \frac{v^2}{r} = 55\text{ m/s}^2$$



- (a) Draw the free body diagrams of both objects.
Teken die vryliggaam-diagram vir beide voorwerpe.



(4)

- (b) Determine the magnitude of the acceleration of the objects. In which direction is this acceleration? Show your choice of coordinate system clearly.
Bepaal die grootte van die versnelling van die voorwerpe. In watter rigting is hierdie versnelling?
Dui jou keuse van koördinaatstelsel duidelik aan.

(5)

Assume acceleration is down the slope ✓

$$\text{For } m_1: \Sigma F_{y m_1} = m_1 a_y = T - m_1 g$$

$$T = m_1 (a_y + g) \quad \text{--- ①}$$

$$\text{For } m_2: \Sigma F_{x m_2} = m_2 a_x = -T + m_2 g \sin \theta$$

Substituting from ①

$$m_2 a_x = -m_1 (a_y + g) + m_2 g \sin \theta$$

$$\text{but } a_x = a_y = a$$

$$\therefore a = \frac{-m_1 g + m_2 g \sin \theta}{m_2 + m_1} \quad \checkmark$$

$$a = \frac{-(2.00 \text{ kg})(9.8 \text{ m/s}^2) + (6.00 \text{ kg})(9.8 \text{ m/s}^2) \sin 55^\circ}{(2.00 \text{ kg} + 6.00 \text{ kg})}$$

$$a = 3.57 \text{ m/s}^2 \quad \checkmark$$

ENJOY THE REST OF THE WEEKEND /
GENIET DIE RES VAN DIE NAWEEK